Priyanshu Kumar Senapati (1RUA24CSE0337)

Owns: Classifier training, calibration, safety metrics, deployment. **Demo slot:** 0:00-0:50: problem and the clear sample.

Your part

Question	Short answer
How was it trained?	ConvNeXt-Small fine-tuned at 384 px on ground-truth crops (box + 8% margin), with flips, darkening, jitter and random erasing. Stratified 90/10 validation split, seed 42. The test split was used once, for final numbers.
Why does the crop help?	Whole photo 91.66%, crop 94.55%: more pixels on the car. That 94.55% uses annotated boxes, so say so; detector plus classifier end to end isn't measured yet.
How is it deployed?	One warm 2-CPU Modal container with FastAPI and ONNX Runtime. Without model files the app falls back to sample answers instead of crashing.

Your failure case: Audi S5 taken for A5 (15 photos, our most frequent error): the trims differ mainly in grille, bumpers and badges.

Everyone must be able to answer

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Why ConvNeXt at 384 px?	Fine-grained classes differ in small details, so resolution matters. ConvNeXt-Small (ImageNet-22k pretrained) beat Tiny on validation, 94.23% to 93.74%. Test: 94.55% top-1, and 96.7% of unknown cars caught against Tiny's 88.3%.
Why fine-tune YOLO?	COCO weights already found the single car (mean IoU 0.886) but boxes were loose. Fine-tuned on the Stanford **train** boxes as one class: IoU 0.954, mAP@0.5 0.999, mAP@0.5:0.95 0.941. No crash data exists to tune on, so that gap remains.
What did CLAHE do?	Alone before the classifier it barely moved dark-photo accuracy (within +/-0.15 points), because training already darkens images. It retries the detector and drives the pipeline view. Dark crops now also get a second reading from a CLAHE copy, averaged with the original: neutral to +0.5 points.
Temperature scaling and ECE?	Divide logits by T = 0.768, fitted on validation, so "80% sure" means about 80% right. ECE fell from 0.098 to 0.011. Accuracy doesn't change, but thresholds become meaningful.
The summed-probability rule?	Add up the probability of all 9 hybrid/EV classes and warn at 0.05 or above. It misses 1.15% with 0.69% false alarms, against 4.32% missed if we trust the top answer. A miss is far worse than a false alarm.
How does the unknown check work?	1 minus the mean cosine similarity to the 5 nearest training embeddings. Threshold: 95th percentile on validation. 58 of 60 unseen cars caught; 8.3% of known cars wrongly flagged.
Why dynamic INT8, MatMul/Gemm only?	ConvNeXt's heavy layers are Linear, and quantising Conv needs ConvInteger, which ONNX Runtime CPU lacks for int8. 189.4 MB to 54.3 MB, 739 ms to 302 ms, no accuracy loss on a 300-image check.
What are the limits?	US-market cars up to 2012, trained on no crashed vehicles. Weak spots: severe blur 40%, very low resolution 53%, heavy occlusion 66%. Close look-alikes fool it: 5 of 8 new Mahindra Thar photos were called a Jeep Wrangler. Always confirm with VIN or plate.
What is the brand expert?	A second ConvNeXt-Tiny trained on zoomed parts (badge, grille, lights) votes over 9 windows, combined as whole^0.8 x close-ups^0.2. Brand accuracy 98.30% to 98.61%; 30% hidden 90.75% to 92.05%.
Real crash photos?	48 from Wikimedia: declined 44, 3 confident, 1 of those wrong. Brand right only 23%. Most are outside the 196 classes, so declining is the intended behaviour.
Is the classical part useful?	Classical alone gets 13.7%. Fusion weight came out 0 on validation, and we report that. Its value: the detector retry, the quality reasons behind low confidence, and the visible pipeline. The hybrid gain is the crop (+2.9 points), calibration and safety rules.

If you don't know: say so, then say where the number lives (`eval_P5_T50.md`). Never guess a number.

Saatwik Kumar Yadav (1RUA24CSE0390)

Owns: Product and UI: live camera, crash handling, browser tests, CI. **Demo slot:** 1:20-2:20: live camera and stress test.

Your part

Question	Short answer
How does the lock work?	One frame every 1.5 s, one request at a time. 3 agreeing frames in a row lock it. For an unknown car the closest match is ignored; for an ambiguous one only the level it is sure of counts.
How is live mode kept safe?	The high-voltage state is part of what must agree, so a flickering warning can never lock as safe. A warning shows on the first frame that raises it and stays for 3 frames.
Is the edge overlay real?	Yes. Sobel plus non-maximum suppression (the core of Canny) runs in the browser about 12 times a second, brightest inside the detector box. It is visual feedback; the answer comes from the server.
What is the automatic sweep?	It re-renders the photo at 8 slider positions of one condition, sends each to the real model and plots usable confidence, marking the first level where the answer changes. Real calls, nothing precomputed.
Is the rescue sheet accurate?	It marks the region the manufacturer's guide describes on a generic body shape. Not to scale, and the app says it isn't a guide for where to cut.
What if something breaks?	An error boundary shows "Start again" plus safe advice. Blocked camera and lost server each have their own message. Browser tests cover these and run on every push.

Your failure case: Mahindra Thar answered as Jeep Wrangler in 5 of 8 new photos, with no high-voltage warning. It is styled on the Wrangler, so its embedding sits close to a known car and the unknown check passes it.

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Rachana Bhaskar Gowda (1RUA24CSE0348)

Owns: Safety content and real-world validation. **Demo slot:** 0:50-1:20: the hybrid that looks like a petrol car.

Your part

Question	Short answer
Where do battery locations come from?	Read from manufacturer guides for all 9 models, with pages listed in docs/hazard-sources.md. Only the most likely hybrid's location is shown, and it names that car.
How far can they be trusted?	Seven links are copies hosted by fire services or rescue sites, the Malibu guide is older than the 2010 car, and the BMW sheet is German. No responder has reviewed the text yet.
Any surprise in the guides?	The Malibu Hybrid's cables are blue and 36-42 V, not orange, so "look for orange" alone would be wrong. The app's advice now says so.
Why does India need this?	4,80,583 accidents and 1,72,890 deaths in 2023 (MoRTH). EV registrations 1.74 lakh in FY20 to 19.68 lakh in FY25 (Lok Sabha, Vahan). Most EVs are two- and three-wheelers, which we don't cover.

Your failure case: Sonata vs Sonata Hybrid: same body, different powertrain. Summing probabilities cuts missed hybrids from 4.3% to 1.2%.

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R Muni Disnitha (1RUA24CSE0345)

Owns: Evidence, documentation and demo. **Demo slot:** 2:20-3:00: Model report and closing line.

Your part

Question	Short answer
Where do your numbers come from?	eval_P5_T50.md: the official 8,041-image test split, never used for tuning. A test fails if the report page disagrees with it.
What was the ablation result?	Remove the crop: 94.55% to 91.66%. Remove classical fusion: no change. Add flip TTA: 94.70%. FP32 and INT8 match on the subset.
How honest is the unknown number?	60 photos is small, and they influenced our choice of model. A later check on 8 new Thar photos caught only 3, so look-alikes are worse than 96.7% suggests. The parking-lot photos are a second, independent test.
What if the demo fails?	A recorded backup video follows the same 3-minute script, kept on two laptops and a phone.

Your failure case: Heavy blur: top-1 falls to 40%, but the "Blurry" check explains why confidence is low instead of hiding it.

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